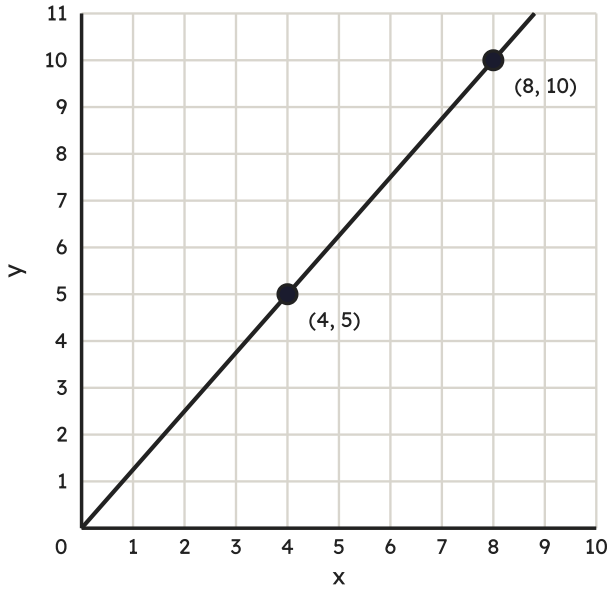
Benchmark: **MA.7.AR.4.4** · Translate between representations Student code: _____ Calculator: yes

Learning goal: I can write an equation for a proportional relationship. (circle your level after scoring)

4	I can translate between a table, a graph, an equation, and a description of the same relationship.
3	I can write the equation y = kx from a table, graph, or description and use it.
2	I can find the constant of proportionality.
1	I need help to start this benchmark.

1. The graph shows a proportional relationship between batches made (x) and cups of flour used (y).

Write an equation for this relationship.



2. A recipe uses $\frac{5}{4}$ cup of sugar for each batch of muffins. The relationship is proportional. Let x be the number of batches and let y be the number of cups of sugar.

Which equation represents this relationship?

A. $y = \left(\frac{5}{4}\right) x$

B. $x = \left(\frac{5}{4}\right) y$

C. $y = x + \frac{5}{4}$

D. $y = \left(\frac{4}{5}\right) x$

Turn over →

Benchmark: **MA.7.AR.4.4** · Translate between representations Student code: _____

3. The table shows a proportional relationship between x and y.

x	y
2	6
5	15
7	21

Write an equation for this relationship.

4. A machine fills 4.5 bottles each minute that it runs, on average. The relationship between the number of minutes and the number of bottles is proportional.

Number of minutes	Number of bottles
6	?
8	?
10	?

What are the three missing values in the table?

5. Diego earns \$11 per hour walking dogs. The relationship between the money earned, e, and the hours worked, h, is proportional.

Which equation gives the number of hours, h, that Diego must work to earn e dollars?

A. $h = 11e$

B. $h = e - 11$

C. $e = \left(\frac{1}{11}\right) h$

D. $h = \left(\frac{1}{11}\right) e$

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.4.4 · MA.7.AR.4.4 · Translate between representations

Unit 11: Representing Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $y = 1.25x$

Use a labeled point: $(4,\ 5)$. The constant of proportionality is $5 \div 4 = 1.25$.
The equation is $y = 1.25x$.

2. Regular **A.** $y = \left(\frac{5}{4} = 1\frac{1}{4}\right)x$

For each one of x, y increases by $\frac{5}{4}$, so the constant of proportionality is $\frac{5}{4}$ and $y = \left(\frac{5}{4}\right)x$.
$y = x + \frac{5}{4}$ would add $\frac{5}{4}$ once; $x = \left(\frac{5}{4}\right)y$ and $y = \left(\frac{4}{5}\right)x$ reverse the rate.

3. Regular $y = 3x$

The constant of proportionality is $y \div x = 6 \div 2 = 3$.
A proportional relationship has the form $y = kx$, so $y = 3x$.

4. Regular **27, 36, 45. Score: 1 point for all three values; two values earn 0.**

Multiply each number of minutes by 4.5, the number of bottles for one minute.
$4.5 \times 6 = 27$
$4.5 \times 8 = 36$
$4.5 \times 10 = 45$

5. Rigor **D.** $h = \left(\frac{1}{11}\right)e$

Each hour earns \$11, so each dollar takes $\frac{1}{11}$ of an hour.
$h = \left(\frac{1}{11}\right)e$. This is the same relationship as $e = 11h$, solved for h.

Re-check strip (3 items, C-AR.4.4-R)

1. Regular $y = 3.75x$

Use a labeled point: $(4,\ 15)$. The constant of proportionality is $15 \div 4 = 3.75$.
The equation is $y = 3.75x$.

2. Regular **A. The printer prints 18 pages each minute.**

In $y = 18x$, the constant 18 is the number of pages for each minute of printing: the printer prints 18 pages each minute.
The relationship is not additive (18 is not added on each time), 18 is not a total, and the rate is pages per minute of printing, not minutes of printing per page.

3. Regular **C.** $y = \left(\frac{1}{2}\right)x$

For each one of x, y increases by $\frac{1}{2}$, so the constant of proportionality is $\frac{1}{2}$ and $y = \left(\frac{1}{2}\right)x$.
$y = x + \frac{1}{2}$ would add $\frac{1}{2}$ once; $x = \left(\frac{1}{2}\right)y$ and $y = 2x$ reverse the rate.

Comparison strip (2 items, C-AR.4.4-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $y = 2.5x$

Use a labeled point: $(4,\ 10)$. The constant of proportionality is $10 \div 4 = 2.5$.
The equation is $y = 2.5x$.

2. Regular **A.** $y = \left(\frac{3}{2} = 1\frac{1}{2}\right)x$

For each one of x, y increases by $\frac{3}{2}$, so the constant of proportionality is $\frac{3}{2}$ and $y = \left(\frac{3}{2}\right)x$.
$y = x + \frac{3}{2}$ would add $\frac{3}{2}$ once; $x = \left(\frac{3}{2}\right)y$ and $y = \left(\frac{2}{3}\right)x$ reverse the rate.

Reteach plan — C-AR.4.4

MA.7.AR.4.4 · Translate between representations

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

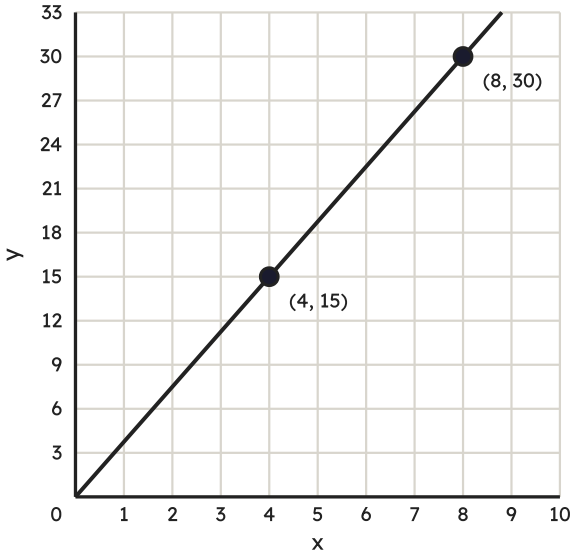
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.4.4-R

Benchmark: **MA.7.AR.4.4** · Translate between representations Student code: _____ Calculator: yes

1. The graph shows a proportional relationship between hours (x) and miles walked (y).

Write an equation for this relationship.



2. A printer prints at a constant rate. The proportional relationship is represented by $y = 18x$, where x is the number of minutes of printing and y is the number of pages.

Which description matches this relationship?

A. The printer prints 18 pages each minute.

B. The printer prints 18 more pages each minute than the minute before.

C. The printer takes 18 minutes for each page.

D. The printer prints 18 pages in all.

3. A gardener uses $\frac{1}{2}$ bag of soil for each planter. The relationship is proportional. Let x be the number of planters and let y be the number of bags of soil.

Which equation represents this relationship?

A. $y = 2x$

B. $x = \left(\frac{1}{2}\right) y$

C. $y = \left(\frac{1}{2}\right) x$

D. $y = x + \frac{1}{2}$

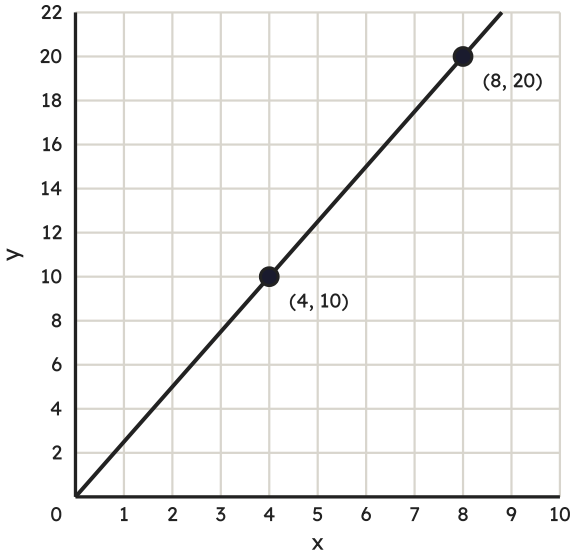
Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-AR.4.4-X

Benchmark: **MA.7.AR.4.4** · Translate between representations Student code: _____ Calculator: yes

1. The graph shows a proportional relationship between minutes (x) and pages read (y).

Write an equation for this relationship.



2. A recipe uses $\frac{3}{2}$ cup of sugar for each batch of muffins. The relationship is proportional. Let x be the number of batches and let y be the number of cups of sugar.

Which equation represents this relationship?

A. $y = \left(\frac{3}{2}\right) x$

B. $y = \left(\frac{2}{3}\right) x$

C. $x = \left(\frac{3}{2}\right) y$

D. $y = x + \frac{3}{2}$

Score: ____ / 2 → copy to your progress strip